Release readiness and grip force regulation during robotic handover to humans

Abstract

Releasing an object requires a robot to decide whether a person can receive it and then reduce its own grip without disrupting the transfer. This focused narrative review compares visual, force and tactile evidence for these decisions. We organize methods by the inference supported by their observations and distinguish release triggering from grip unloading. Visual enclosure, pulling force and reduced robot load provide different, incomplete evidence of readiness; temporal contact patterns and active probing address additional ambiguities. Within-study comparisons support complementary sensing and coordinated unloading. Their different success endpoints, however, prevent a universal ranking. Object mass, transfer direction, receiver instructions and actuator delay materially affect interpretation. We propose a reporting framework connecting evidence, decision and unloading, with separate measures of premature release, missed release, latency and interaction effort. The framework supports more informative comparisons without requiring new participant data.

Keywords: robot-to-human handover; object release; grip force control; tactile sensing; force sensing; sensor fusion

1 Introduction

A robot can detect a hand touching an object before that hand can safely support it. Conversely, waiting for a large pull can prolong an otherwise secure exchange. Existing reviews organize handovers by interaction phase, robot role, end effector or capability, and summarize human coordination.[1,2,3] This review examines the narrower transition from shared contact to complete robot release. It asks what each measurement establishes about receiving readiness, how that judgment drives unloading, and which experimental conditions make reported outcomes comparable.

The contribution is an evidence, decision and unloading framework linking sensing to the claim it supports, together with an endpoint audit and a condition-based reporting scheme. This is an analytical synthesis, not a validated controller. Human experiments inform design; their results are not treated as robot performance.

2 Review approach

Targeted web searches and citation following were conducted on 19 October 2026 using combinations of handover, release, grip force, vision, force, tactile, weight and disturbance. Publisher pages, Crossref records, author repositories and arXiv were checked. Selection prioritized studies of robot giver release, relevant human force studies and recent reviews. Planning-only and human-to-robot studies were used as context. This is a focused narrative review without exhaustive database coverage, independent screening or meta-analysis. Search queries, source access levels, extraction locators and discrepancies accompany Supplementary Material S1 and the evidence spreadsheet. SANRA informed transparency

checks, without constituting a robotics reporting standard.[4] Numerical comparisons below require accessible full text; abstract-only records support limited qualitative statements.

3 From observed contact to justified release

Figure 1 separates observation, inference and execution. Readiness has at least three distinguishable components: contact with the object, an action consistent with taking it, and sufficient receiving support. None automatically establishes the others. The framework groups evidence by its inferential role rather than declaring one sensor universally superior (Table 1).

External vision observes approach and hand enclosure. Wrist or joint sensing measures aggregate interaction; fingertip sensing resolves local contacts. Optical tactile sensors belong to the latter category even though an internal camera reads deformation.[5] Sensor technology and inference should therefore be coded separately. The same wrist sensor can support a magnitude threshold, a temporal classifier or an actively identified contact model.

Grip force denotes contact normal force; load force denotes the supported gravitational and dynamic load. During approximately stationary transfer, robot and human vertical support forces together balance object weight. Robot unloading is therefore informative, but an upward collision can mimic reduced load. Horizontal pulling and torsional disturbances require different interpretation. During arm motion, gravity and inertial compensation help separate receiver forces from loads generated by the robot.[6]

Readiness inference is also distinct from execution. Opening fingers, tracking a release profile and continuously regulating grip force provide different opportunities to correct an uncertain transfer. A successful binary trigger alone does not validate continuous grip control.

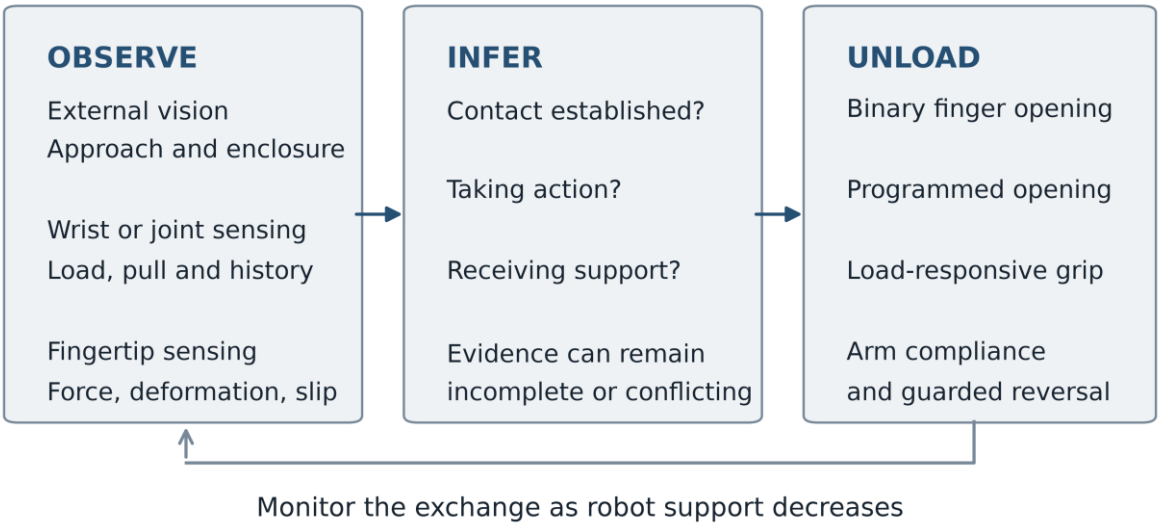


Figure 1. Proposed analytical framework linking observations, readiness inference and unloading. A measurement may support more than one inference, and uncertainty can persist during unloading. The diagram is a synthesis of the reviewed mechanisms, not a validated control architecture.

Table 1. Evidence classes and their limits. Classes may be combined; they are not a performance hierarchy.

Evidence	Supports	Remaining ambiguity
Approach or enclosure	Contact opportunity	Visible fingers may not support load
Passive load or pull	Load sharing or taking action	Bumps, direction and inertia confound interpretation
Temporal contact pattern	Sustained action under a learned protocol	Unseen users, objects or action histories
Actively probed response	Model-based grasp firmness	Model assumptions and untested disturbances

4 Comparing sensing and unloading strategies

Visual prediction helps prepare a release. Controzzi et al. found that human passers with visual access began releasing around receiver contact and adjusted release rate to reaching speed; removing vision delayed release.[7] These findings motivate anticipation, not a claim that robot vision alone measures receiving support. Mohandes et al. combined visual fingertip enclosure with a joint torque action classifier. Their within-study comparison favored fusion over either stream (Table 2), showing the value of testing complementary failure modes.[8]

Load-responsive control links unloading to the physical exchange. Chan et al. implemented grip regulation from measured load on a PR2.[9] Medina et al. coupled arm motion and grip regulation through estimated load share, reporting reduced internal forces and shorter passing times than a switching baseline.[10] Their robot evaluation used a 0.556 kg bottle; the result does not establish performance across arbitrary masses. Costanzo et al. required load sharing before a pulling cue could complete release and demonstrated renewed holding when the receiver withdrew.[11] Their main robot-to-human examples used objects of 0.290, 0.103 and 0.082 kg. These demonstrations support the control mechanism rather than a population success estimate. Receiver-guided motion offers another convention, allowing the person to redirect the object before release.[12]

Tactile feedback can distinguish disturbances while maintaining robot grasp. Gómez Eguíluz et al. combined fingertip force adaptation with direction-gated release. Increasing the persistence requirement traded responsiveness for fewer false classifications; the reported zero false positives applies to the evaluated data, not unrestricted handovers.[13] Learned wrench histories offer another route: Khanna et al. also evaluated a release technique learned from human exchanges in robot-to-human transfers.[14] Temporal prediction should be distinguished from direct measurement of a secure receiving grasp.

Penzotti and Controzzi separated contact detection, command latency and finger opening. Their proprioceptive compensation and adaptive release profile addressed handovers during arm motion.[6] The object-mounted contact sensors measured reference events for evaluation; they did not trigger the controller. Adaptive opening is consequently not interchangeable with closed-loop regulation of normal force.

Recent preprints extend readiness inference. Li et al. actively probed contact and checked model-predicted support and opposing force responses. Their blindfolded receiver study favored probing over force and load thresholds.[15] All conditions used the same visual guidance. Vertical probing and rigid objects delimit the result; lateral pushes remain a stated confound. Marra et al. combined temporal tactile encoding with a learned visuomotor policy and compliance.[16] Ablations favored the combined configuration, but success mixed hold-expected and release-expected trials. Its timing-constrained outcome was different again. These findings justify further testing of contact history and compliance, not a general safety guarantee.

Table 2. Selected results within their original conditions. Values are author-reported; percentages across rows have different meanings.

Study	Evaluation conditions	Result and endpoint
Mohandes et al.[8]	30 volunteers; six scripted actions; 180 trials per method; one test object, mass unspecified in examined text	Correct hold/release: vision 79%, torque 90%, fusion 98%; fusion 177/180. Source count discrepancy documented in S1
Penzotti and Controzzi[6]	15 participants; prism <0.100 kg; two approach speeds; 36 valid repetitions each	Contact to detection: 93 ± 43 ms at low speed; 72 ± 43 ms at high speed; not total transfer time
Li et al.[15]	Preprint; 12 blindfolded receivers; 30 rigid objects, 10 per person	Firm grasp at release: active 97.5%, force threshold 67.5%, load threshold 65.0%
Marra et al.[16]	Preprint; 10 receivers; bimanual box; 100 trials per policy, half hold-expected	Correct trial outcome: full 93%, no tactile 54%, no compliance 82%; full median release delay 1.70 s

5 Why conditions change the comparison

Mass affects both mechanics and observability. For a light object, force sensor uncertainty may approach its weight; for a heavy object, a fixed pull threshold represents a smaller relative effort. A useful diagnostic is $\sigma_F/(mg)$, the ratio of force measurement uncertainty to object weight. Here, σ_F denotes force uncertainty, m object mass and g gravitational acceleration. This ratio describes measurement difficulty; it is not a validated release threshold. Reporting mass, sensor uncertainty, grip geometry and motion compensation together makes the limitation assessable. The YCB-Handovers dataset offers weight-varied human motion data, but this does not itself validate a robot release trigger.[17] Supplementary S1 distinguishes it from release experiments and from related, overlapping preprints.

Direction must specify object orientation, extraction motion and the gravity axis. An upward pull changes support and communicates taking simultaneously; a horizontal pull may communicate taking without removing vertical load. Direction-gated tactile results therefore cannot be transferred unchanged to another interaction convention.

Evaluation also determines what success means. Correctly retaining an object after a bump, completing a transfer and being judged firmly grasped at release answer different questions. Penzotti and Controzzi repeated invalid trials; Li et al. reduced visual compensation through blindfolding; Marra et al. tested deliberately misleading contact. Such protocols expose different strengths and preclude pooling success rates in this review. Repeated trials from one person are not independent participants, and subjective fluency does not establish low drop risk. Source inconsistencies and unreported conditions must remain visible rather than being resolved through assumptions.

6 Implications and reporting priorities

Controller selection should begin with the ambiguity that must be resolved. Approach prediction can prepare a release, load sensing can guide unloading, and contact history or probing can test competing explanations. The appropriate combination depends on object mechanics, sensing access and actuator authority. More sensing without a matching decision rule need not improve reliability.

We recommend reporting four event times: first receiver contact, release decision, actual finger motion and final robot separation. Their differences distinguish inference delay from communication, actuation and unloading. Report premature releases on hold-expected trials and missed or delayed releases on take-expected trials separately, with denominators and per-participant variation. Add peak interaction force, mass, pull direction, receiver visibility, disturbance protocol and stopping rules. Supplementary Figure S1 provides an event template. Future benchmarks should vary weight independently of shape and compare ordinary receiving with misleading contacts under matched conditions.

This synthesis is limited by targeted retrieval, access restrictions, heterogeneous endpoints and two recent preprints. It does not estimate failure probabilities in deployment or validate the proposed framework. Its practical conclusion is that release timing and grip unloading should be assessed together, while preserving the distinct evidence needed to justify each decision.

7 Conclusion

Robot release requires a justified transition from observed interaction to receiving support and coordinated unloading. The reviewed evidence favors evaluating complementary cues and feedback within explicit conditions. Reporting event times, failure types and physical context would make future comparisons more useful than ranking incompatible success rates.

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